

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 40)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain the locational requirements for harnessing wave power.

[2 marks]

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0 1 . 2

Describe how **one** named wave power device generates electricity.

[2 marks]

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0 1 . 3

Give **two** design problems that have delayed the development of wave power.

[2 marks]

1

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2

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0 1 . 4

Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

[4 marks]

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10

- 0 2** Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD ₅₀ for pest / mg kg ⁻¹	LD ₅₀ for ground beetle / mg kg ⁻¹	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

- 0 2** . **1** Explain what is meant by LD₅₀.

[1 mark]

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- 0 2** . **2** Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]

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- 0 2** . **3** Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]

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- 0 2** . **4** Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

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0 2 . 5

Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

1

2

10

0 3

A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm ⁻³	River flow / m ³ s ⁻¹
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 3 . 1

Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer

 %

0 3 . 2

Explain the large decrease in concentration between sites C and D.

[2 marks]

03 . 3

Explain why the concentration changes very little between sites D and E.

[2 marks]

03 . 4

Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

03 . 5

Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

10

0 4 . 1

Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 4 . 2

Describe **four** principles of designing vehicles for end of life.

[4 marks]

1

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4

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0 4 . 3

Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 4 . 4

Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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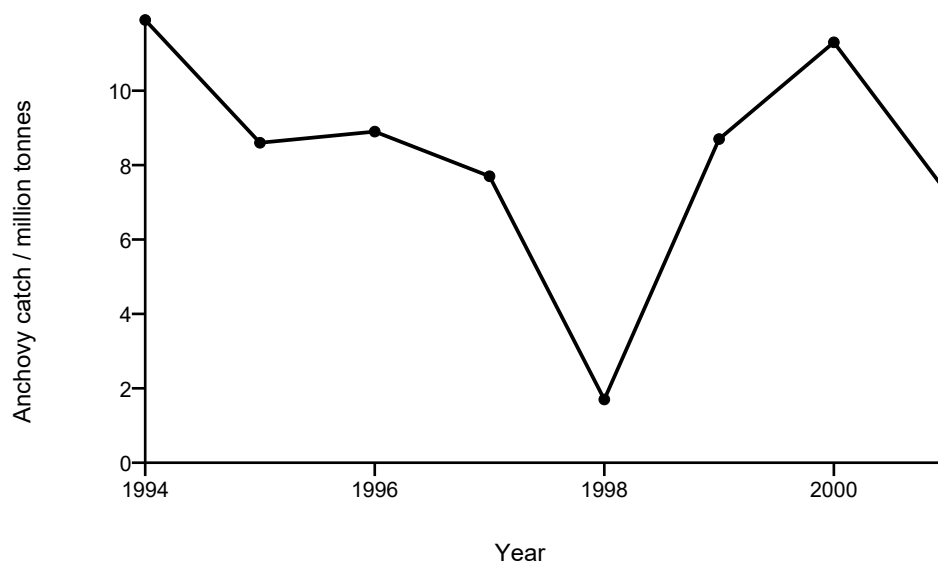
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0 5

Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



0 5

1

Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 5

2

Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

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0 5 . 3

Describe **two** other effects of El Niño events.

[2 marks]

1

2

0 5 . 4

Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

10

0 6 . 1

Describe how soil texture is determined using a stack of sieves.

[3 marks]

0 6 . 2

Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

06 . 3

Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

06 . 4

Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

10

07

In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

07 . 1

Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

07 . 2

Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

07 . 3

Compare the toxicity of the three chemical forms of mercury.

[3 marks]

10

0 8 . 1

Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen

☐

B Oxygen

☐

C Argon

☐

D Carbon dioxide

☐**0 8 . 2**

Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 8 . 3

Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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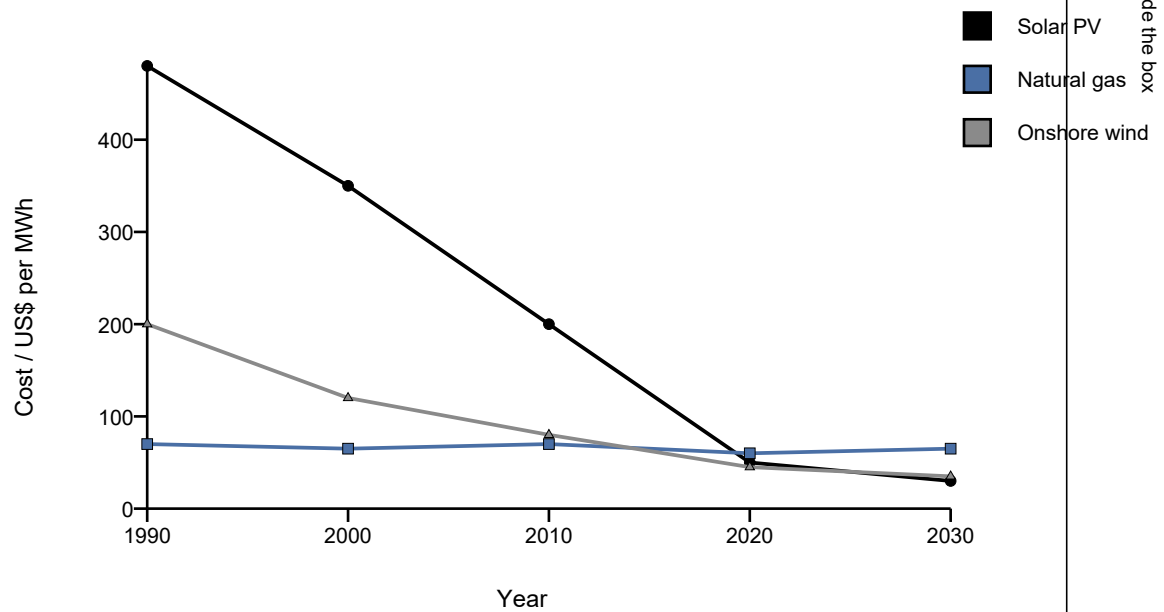
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09

Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



09

1

Describe the trends shown in Figure 1.

[3 marks]

09

2

Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

09 . 3

Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

09 . 4

Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

10

10 . 1

Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

Do not write outside the box

Extra space

10 . 2

State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS